

Recent Advances in Animal Models, Diagnosis, and Treatment of Temporomandibular Joint Osteoarthritis

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Abstract

Temporomandibular joint osteoarthritis (TMJOA) is a gradual degenerative jaw joint condition. Until recent years, TMJOA is still relatively unrecognized and ineffective to be treated. Appropriate animal models with reliable detection methods can help researchers understand the pathophysiology of TMJOA and find therapeutic options. Here, we summarized common animal models of TMJOA created by chemical, surgical, mechanical, and gentical appraoches. The relevant pathological symptoms and induction mechanisms were outlined. In addition, different pathological indicators Furthermore, emerging therapeutic regimens, such as intra-articular drug delivery and tissue engineering-based approaches to treat TMJOA based on these animal models, were summarized and updated. Understanding the physiology and pathogenesis of the TMJOA, together using various ways to diagnose the TMJOA were elaborated, including imaging techniques, molecular techniques for detecting inflammatory cytokines, histochemical staining and histomorphometry measures.with a more reliable diagnosis, will enable the development of new prevention and more effective treatment strategies and thereby improve the life quality of TMJOA patients.

Keywords

Temporomandibular joint osteoarthritis, Animal model, Diagnose, Treatment, New methods

1. Introduction

The temporomandibular joint (TMJ) is involved in mastication, swallowing, breathing, speaking, and other common activities of daily living. TMJ osteoarthritis (OA) is a gradual degenerative condition in the jaw joint, and affects 8 to 16 percent of the population worldwide, with an annual health expense of more than \$80 billion.^{1,2} Orofacial pain, joint noises, and limited jaw movement are typical symptoms.

The typical pathological changes of TMJOA are progressive cartilage deterioration, subchondral bone remodeling, and synovitis. The etiology of TMJOA is complex and multifactorial. The root cause of TMJOA is assumed to be mechanical overload on joint components that surpasses physiological resistance, such as excessive loading on healthy articular cartilage or normal loading on the damaged articular cartilage, which disrupts cartilage matrix homeostasis and leads to TMJOA.^{3,4}

Although hundreds of researches have looked into the mechanisms that cause TMJOA, it is still poorly understood due to various pathogenic variables and clinical presentations. Different animal models have been developed to study the pathological causes of TMJOA. These models help elucidate the mechanics of TMJOA and evaluate therapeutic efficacy even though they cannot reflect all clinical features.⁵

A reliable diagnosis of TMJOA is recognized as essential to confirm the successful establishment of relevant animal models and follow-up treatment. The radiographic characteristics of the condyle or articular eminences, such as bone erosive degradation, osteophyte formation, and cyst-like alterations, are frequently used to diagnose TMJOA in the clinic.^{6,7} Pathological diagnosis of TMJOA based on histochemical staining and histomorphometry measurements and molecular techniques for detecting various inflammatory factors were developed recently to assess the success of animal models and the efficacy of novel therapeutic approaches.

Several treatments are currently available to alleviate symptoms of TMJOA.⁸ These therapies efficiently reduce signs and symptoms in most cases, but their therapy effect on the degenerative cartilage and subchondral bone is insufficient. Emerging therapeutic

techniques, such as innovative intra-TMJ drug delivery and tissue engineering-based products, are being investigated as an alternative therapeutic approach to treat TMJOA.

In this review, we firstly introduce the currently established animal models of TMJOA. The diverse diagnosis approaches to TMJOA and their pathological mechanisms are summarized. The reported treatment strategies for TMJOA based on these animal models are also elaborated. Further, since current animal models, diagnosis, and clinical treatments are still in their infancy, we point out their limits and future strategies to provide an up-to-date review.

2. Animal Models of TMJOA

Animal models of TMJOA are essential for researching the pathophysiology and testing the efficacy of novel therapy options.⁹ Various attempts have been carried out to produce TMJOA animal models, including naturally occurring, chemical induction, surgical operation, mechanical stimulation, and genetically modified models. In this section, we summarized the different TMJOA animal models according to the different causes of TMJOA (Table 1& Figure1).

2.1 Naturally occurring models of TMJOA

As a degenerative disease, the prevalence of TMJOA rises with age. Zheng et al¹⁰ found a naturally occurring TMJOA model using aging mice. In their investigation, 60-week-old mice showed degeneration and decreased bone volume in cartilage and more TRAP-positive cells in the subchondral bone. Besides, Ukita et al¹¹ also found exposed proliferative layer in the damaged TMJ cartilage of the 20-month-old mice. Using the aging mice with spontaneous TMJOA can help researchers to study the pathological relationship between aging and cartilage and subchondral bone lesions in TMJOA.

2.2 Chemical Induction of TMJOA

Chemical stimulants including complete Freund's adjuvant (CFA), monosodium iodine acetate (MIA), vascular endothelial growth factor (VEGF), collagenase, and formalin, are commonly used to create TMJ inflammatory models. Most chemical induction methods centered on the inflammatory part of TMJOA. They appear to be straightforward to

employ, with predictable damage and repeatability, but these attempts are mainly applied on rats and only replicate a single symptom of TMJOA.⁵

2.2.1 Chemical Induction by CFA and MIA

CFA contains heat-killed mycobacterium tuberculosis, which is thought to produce acute inflammation. MIA causes chondrocyte death by decreasing the activity of glyceraldehyde-3-phosphate dehydrogenase. Cui et al¹² investigated the alterations of condylar cartilage and subchondral bone in CFA-induced inflammation in rats. Reduced cartilage thickness, fewer chondrocytes, and decreased proteoglycan expression were found in the articular cartilage of inflammatory TMJs. The RANKL/OPG ratios in the cartilage and subchondral bone were also found to increase during the CFA injection. Jiang et al injected 50 μ l of MIA (4 mg/kg) into the upper compartment of the TMJ of rats. They discovered that the rats' condylar structure degraded over time, decreasing chondrocyte quantity and percentage of proteoglycan area in cartilage.¹³ Wang et al investigated rats after injecting different doses of MIA for 12 weeks. They discovered that injecting 0.5 mg or more MIA into the TMJ resulted in OA-like lesions in 4 weeks. Condylar breakdown progressed throughout time, beginning with chondrocyte death and progressing through cartilage matrix disorganization, subchondral bone erosion, and osteophyte development.¹⁴ By changing its concentration, the MIA-induced model of TMJOA has the benefit of being simple to modify the severity of the joint lesions.

2.2.2 Chemical Induction by VEGF and collagenase

VEGF is a pro-angiogenic growth factor and implicated in the breakdown of articular cartilage and the progression of TMJOA by activating osteoclasts. Subchondral bone destruction, noticeable cartilage degradation, and subchondral bone resorption with a substantial number of osteoclasts were found in VEGF-injected mice. The results indicate that VEGF can cause TMJOA by causing cartilage and subchondral bone destruction.¹⁵ TMJOA was also generated by intra-articular injection of reagents to degrade or inactivate collagen. This model was induced by injecting 12.5 μ l of collagenase into the upper cavity of mice TMJ. Li et al. discovered that the mice's joints had an uneven surface, and inflammation increased dramatically.¹⁶ In a rat model, formalin has also been used to

cause the TMJOA. Goulart et al. observed synovial membrane hyperplasia, inflammatory reaction in the condylar neck, and atrophy of the fibrocartilage in the formalin-injected rats.¹⁷

2.3 Surgical Operation of TMJOA

Disc perforation, disc displacement, discectomy and condylectomy are surgical procedures for modeling TMJOA. Rather than rats and mice, larger animals such as rabbits, lambs, and monkeys are used in these animal models.

2.3.1 TMJOA by Disc Perforation

The rabbit is frequently used as a disc perforation model. Embree et al. produced a 2.5 mm defect in rabbits via a unilateral TMJ disc puncture operation. They discovered condylar cartilage degeneration and heterotopic ossification in experimental condyles after disc perforation. Their findings showed that disc perforation could result in TMJOA and aberrant mandibular development.¹⁸

2.3.2 TMJOA by Disc Deformation

Extra-articular surgery was used by Liu et al to unilaterally move the rabbits' jaw in a posterosuperior direction, resulting in joint disc deformation. Synovial hyperplasia and adhesion in joints, as well as a significant deformation of the joint disc, were discovered in the early postoperative phase. The authors concluded TMJOA by this method recapitulated typical OA pathological manifestations.¹⁹

2.3.3 TMJOA by Disc Displacement

Anterior disc displacement is widely recognized as one of the causes of TMJOA, and it is used in animal surgeries as a surgical model. Adult rabbits were surgically induced with anterior disc displacement by Xu et al. After exposing the zygomatic arch suture, a hole was created under the anterior of the zygomatic arch. The disc's anterior portion was then sewed to the hole. In this way, increased chondrocyte death resulting in a reduction in cell density was observed after surgical disc displacement.²⁰

2.3.4 TMJOA by Discectomy

Furthermore, a partial discectomy was performed on one side of the TMJs to create TMJOA mice. Cohen et al²¹ found degeneration in articular cartilage after 4 weeks. Reduced proteoglycan at 12 weeks and articular cartilage loss at 16 weeks were observed. These findings revealed that the mice with partial discectomy led to early-onset TMJOA-like articular cartilage degeneration.

2.3.5 TMJOA by Condylectomy

Condylectomy is also a surgical procedure to induce TMJOA. Vapniarsky et al²² performed condylectomy in minipigs, they found the operation induced subchondral cysts and large osteophytes formation with the extensive irregular cartilage surface. Besides, Gomez et al²³ used the same way to create TMJOA mice. They revealed that condylectomy caused discontinuous cartilage and bone loss, without cartilage regeneration and extracellular matrix formation.

2.4 Mechanical Stimulation of TMJOA

It has been proven that disrupted occlusion can cause a masticatory system imbalance, which can be an approach to create TMJOA. Bite raising and malocclusion are two common ways in which altered loading can be generated in animal models. These methods are affordable, efficient, and repeatable to induce TMJOA mice models (Table 2).

2.4.1 Stimulation of TMJOA by Malocclusion

Another prevalent method to create TMJOA is unbalanced malocclusion. Zhang et al²⁴ and Sun et al²⁵ bonded a metal tube to the rat's left maxillary incisor and another metal tube to the left mandibular incisor to create an aberrant dental occlusion. Meanwhile, Jiao et al²⁶ mesially relocated the first and third molars of the left maxilla and distally relocated the first and third molars of the right mandible. These two models led to the subchondral bone and cartilage deterioration in the mandibular condyle and increased chondrocyte death. These two models can also cause chondrocyte mortality and OA-like lesions in rat TMJ cartilage.

2.4.2 Stimulation of TMJOA by Bite Raising

Given that the bite raising can lead to TMJOA, Matias et al and Jung et al have developed two different models. By placing a U-shaped stainless steel wire on the molar teeth, Matias et al²⁷ established a bite-raising animal model. Meanwhile, Jung et al²⁸ developed an animal model by building up resin on the left maxillary molars of rats. Both of them observed the development of TMJOA with a decreased thickness of the condylar cartilage and increased apoptosis in the cartilage and subchondral bone.

2.5 Gene Modification of TMJOA

One major step forward in creating TMJOA animal models was the development of gene editing techniques, which allowed for targeted modification of the animal's genome. These animals are useful for investigating the function of a specific gene and its interaction with other components in the temporomandibular joint. According to a recent study, the short stature homeobox 2 (Shox2) gene plays a crucial role in TMJ development. TMJ defects and premature articular disc erosion, symptoms known as TMJOA, are shown in Shox2-depleted mice^[29]. Besides, Proteoglycan 4 (Prg4)-null mice model³⁰ showed abnormal cartilage, osteophytes formation in the glenoid fossa. Discoidin domain receptor (DDR)-1 knockout mice³¹ had bone surface fissures, proteoglycan degradation, chondrocyte cluster development, and other classic TMJOA histology findings.

3. Diagnosis of TMJOA

It is critical to use various pathological indexes to diagnose TMJOA for assessing the efficacy of TMJOA models and evaluating the efficacy of innovative treatments. Imaging techniques including micro-CT and MRI, molecular techniques to detect inflammatory cytokines, histochemical staining, and histomorphometry measures are the most used diagnosis approaches in animal studies.

3.1 Diagnosis of TMJOA by Imaging Techniques

Unmineralized new bones with the abnormal trabecular bone structure, unusual condylar form, and bone mineral density (BMD) were generally found in the TMJOA condylar cartilage. Computed tomography (Micro-CT) is the most appropriate diagnostic method for

bone structure because it can provide anatomical detail.³² Micro-CT can measure the bone changes in the TMJOA models^{12,14} (Figure 2A). The trabecular microstructure can be assessed using metrics such as bone volume fraction (BV/TV), BMD, trabecular thickness (Tb. Th), trabecular number (Tb. N), and trabecular separation (Tb. Sp). In an investigation by Xu et al³³, more extensive marrow cavities were detected in the subchondral bone two weeks later in the TMJOA animals. BV/TV, BMD, Tb. N, Tb. Th all reduced while Tb. Sp rose in TMJOA groups compared to controls according to micro-CT analysis. Besides, joint internal structure derangement, disc displacement, and joint inflammatory fluid frequently occurred in the TMJOA.³⁴ Magnetic resonance imaging (MRI) is considered to be accurate for diagnosing the morphology and position of TMJ discs and soft tissues.³⁵ Badylak et al found joint internal structure derangement and decreased joint space diagnosed by MRI in the pig TMJOA model induced by discectomy.³⁶ With the aid of MRI, Chan et al. found bone degeneration and joint inflammatory fluid in rabbits injured TMJs induced by discectomy³⁷, and Sun et al, found disc displacement in the pig TMJOA models induced by condylar fracture³⁸ (Figure 2B). Compared to Micro-CT, MRI can detect early disc pathological changes³⁹, and display discs, cartilages, and other soft tissues much more clearly in the diagnosis of TMJOA, for its superior resolution and great tissue contrast.^{40,41} However, given its high acquisition cost and a long examination time⁴², MRI is more frequently used in clinically diagnosing TMJOA rather than in animal studies.

3.2 Diagnosis of TMJOA by Molecular Techniques to Detect Inflammatory Cytokines

Inflammation plays a vital role in the development of TMJOA, particularly the tumor necrosis factor α (TNF- α) and pro-inflammatory factors Interleukin-1 β (IL-1 β) and regulate articular cartilage matrix degeneration. Several other cytokines such as IL-6, IL-15, and IL-21 have also been shown to be implicated in TMJOA and can be quantified by ELISA. In some studies⁴³⁻⁴⁵, cytokines levels and gene expression of matrix degrading proteases in the TMJOA animal models were assessed using real-time PCR and western Blot (Figure 2C-D). Total RNA was extracted from condylar cartilage and subchondral bone. The concomitant pro-inflammatory cytokines such as TNF- α , IL-1 β , and several other cytokines were found to be up-regulated in TMJOA. Inflammation causes an increase in these cytokines and leads to degenerative effects.⁴⁶⁻⁴⁹ By the same method, increased

concentration of monocyte chemoattractant protein (MCP)-1 and carboxy-terminal telopeptides (CTX)-1 were also observed in the synovial tissues and fluids of models with TMJOA^[50, 51]. Besides, several studies have discovered the receptor activator of NF- κ B ligand (RANKL) and osteoprotegerin (OPG) play critical roles in the onset and progression of TMJOA. Reduced OPG/RANKL ratios in synovial tissues were found in the TMJOA rats.⁵²

3.3 Diagnosis of TMJOA by Histochemical Staining and Histomorphometry Measures

According to numerous research, synovitis, lower cartilage thickness, fewer chondrocytes, and down-regulated proteoglycan expression have all been reported in the condylar cartilage in TMJOA models. Increased osteoclast activity and larger bone marrow cavity were found in the inflammatory TMJ, with immature new bones, uneven trabecular bone structure, and unusual condylar shape. These histomorphological changes can be measured by histological stainings.

Hematoxylin & Eosin (HE) staining revealed that in TMJOA models, synovial cells had evident hyperplasia, and the inflammatory synovium contained numerous mononuclear cells and lipid droplets. In addition, the number of chondrocytes and the thickness of the condylar cartilage were decreased.³³ Proteoglycan alterations in the cartilage matrix can be determined by T.B. and S.O staining, and proteoglycan amounts were all significantly reduced in the TMJOA models^{18,14} (Figure 2E). The change in trabecula bone shape and the degree of bone mineralization were observed using Masson trichrome staining^[53]. Through it, irregular and disordered trabecular bone structures and large marrow cavities can be observed in the TMJOA models.¹⁴ The osteoclast activity was estimated by tartrate-resistant acid phosphatase (TRAP) staining^{33,54}, and the chondrocytes apoptosis can also be examined in situ using a TUNEL assay.⁵⁵ Wang et al⁵⁶ reported that following TMJOA induction, TUNEL-positive chondrocytes were identified diffusely in the corresponding region. In addition, the upregulation of cartilage degradation markers involved in the pathology of TMJOA such as VEGF, matrix metalloproteinase (MMP)-3, MMP-9, and bone morphogenetic protein (BMP)-7, can be measured by histochemical stainings in the cartilage matrix of TMJOA⁵⁷⁻⁶¹ (Figure 2F).

4. Conservative Treatment of TMJOA

TMJOA treatment aims to inhibit cartilage and subchondral bone deterioration, stimulate condyle remodeling, reduce joint discomfort, and restore TMJ functions.⁶² Manual movement of the TMJ⁶³ and splints have been frequently applied to relieve muscular strain, rectify mandibular malalignment, and reduce pain.^{64,65} Pharmacological treatments such as NSAIDs, muscle relaxants, corticosteroids, and antidepressants are also used to treat TMJOA pain. However, they are commonly connected to side effects such as nausea and dizziness.⁶⁶ If conservative therapy fails, simple non-surgical therapies such as arthrocentesis, arthroscopy, and hyaluronic acid (HA) injections might be used to alleviate TMJOA symptoms.⁶⁷⁻⁶⁹ HA can restore the viscoelasticity of the synovial fluid in an inflamed joint and has lubrication, shock absorption, and joint protection effects.⁷⁰ However, the half-life of HA is short, leading difficulty in sustaining a long-term therapeutic efficacy.

5. Advanced Treatment of TMJOA

Although the conserved treatments listed above can slow disease progression to some extent, they are unable to completely heal the degraded cartilage or subchondral bone defects, as well as degenerative disc disease. These lesions frequently cause severe symptoms, including pain, edema, and dysfunction. Aiming this, novel drug delivery systems and tissue engineering therapies to restore the TMJ tissues gained increasing attention.

5.1 Treatment of TMJOA By Novel Delivery Systems of Therapeutic Agents

The half-life of most intra-TMJ injectable medications in the tissue is short, causing rapid clearance from the site of injection and multiple injections to boost the therapeutic effects. Controlled-release drug delivery systems can address these difficulties by providing medications in a regulated spatiotemporal manner which have been designed for the TMJOA treatments.

Due to the similarity to human tissue structure, biodegradability, and injectability, hydrogels have become the mainly applied materials for the local delivery of drugs to treat

TMJOA. Talaat et al. created chitosan hydrogels with varying β -glycerophosphate (β -GP) concentrations to deliver hyaluronic acid (HA). The hydrogel continuously released HA in rabbit TMJ for 7 days.⁷¹ Moreover, encapsulation of drugs into lipid nanoparticles is able to prolong the delivery and improve the stability of liposoluble drugs. Guilherme et al created a lipid carrier with Pluronic F68 and a blend of cetyl palmitate (CP) and capric/caprylic triglycerides (CCT). This delivery system realized the release of naproxen in the rat TMJ for more than 7 days, which significantly reduced the secretion of inflammatory cytokines IL-1 β and TNF- α .⁷² Nanoparticles with good biocompatibility, biodegradability, and high loading capacity can be elaborated to deliver different proteins, nucleic acids, and chemical drugs. For example, Li et al recently produced mesoporous silica nanoparticles and used them to deliver hyaluronan synthase type 2 enzyme (HAS2) into rat TMJ synoviocytes.⁷³ As a result, endogenous HA synthesis was increased and synovial inflammation was inhibited for more than three weeks with one-shot administration. A similar nanocarrier system was developed in a polymeric micelle manner. Hanafy et al created a micellar system containing PL407, HA, and the anti-inflammatory drug diclofenac potassium (DK).⁷⁴ This formulation showed a controlled DK release for 4 days and regenerated cartilage in OA rats.

5.2 Treatment of TMJOA By Tissue Engineering Strategies

Due to the low regenerative capacity of the condyle, current treatments for TMJOA are mostly symptomatic relief targeted and lack long-term efficacy. Lately, there has been a surging interest in cellular therapy for severe kinds of TMJOA, such as disc thinning, disc perforation, and condyle osteochondral destruction. Bioactive components, including cells and growth factors, are commonly used in such regeneration therapy systems.⁷⁵⁻⁷⁷

5.2.1 Cells in Tissue Engineering Strategies for TMJOA

In TMJ tissue engineering strategies, mesenchymal stem cells (MSCs) are the mostly employed cell type.^{78, 79} Bone marrow is currently the most common MSCs source and bone marrow MSCs (BMMSCs) have been extensively researched. Injection of BMMSCs was found to induce repair and regeneration of TMJOA. Gomez et al revealed that the new cartilage and bone covered the top of the condyle defect with extracellular matrix

deposition in the mandibular condyle cartilage defected mice after 6-week of BMMSCs injection.²³ Aside from bone marrow, adipose tissue is a more accessible source of stem cells. Adipose-derived stem cells (ADMSCs) are more readily available than BMMSCs because of the abundance, high proliferation capacity and differentiation potential.^{80, 81} Ahtiainen et al. implanted polylactide discs, embedded with ADMSCs to the rabbits' defective TMJ discs.⁸² They found ADMSCs promoted cartilaginous tissue formation and bone regeneration in the TMJ. Moreover, a recently published study in 2020 reported that stromal vascular fraction (SVF), a component of lipoaspirate that contained 32% ADMSCs, is more feasible in cellular therapy for TMJOA.⁸³ The SVF reduced inflammatory cytokines, released by the synoviocytes from TMJOA tissue. Given the minimal manipulation and low cost⁸⁴, SVF is worth being more investigated as a new cell source to treat TMJOA in the future. Besides, mesenchymal stem cells from oral tissues, including dental pulp stem cells (DPSCs), periodontal ligament stem cells (PDLSCs), and deciduous teeth stem cells (DTSCs), have been proposed as promising cells in the regeneration systems.^{12,55,85} In an investigation by Cui et al, they injected the DPSCs into the articular cavity of the rat to treat the TMJOA induced by CFA injection. They found that local injection of DPSCs can relieve inflammatory pain, decrease cartilage degradation, induce bone regeneration and downregulate the expression levels of TNF- α , MMP3, and MMP1.¹² Induced pluripotent stem cells (iPSCs) are additional possible stem cell sources. iPSCs have potential in regenerative medicine because of their ability to be induced into various cell lineages including chondrocytes and they are non-invasively harvested and can be transplanted autologously without immune rejection.⁸⁶ Although the use of iPSCs in TMJ disc tissue engineering has not been established, Rim et al. proved the therapeutic effect of iPSC-derived chondrocytes in the osteochondral defect rat models. After 8 weeks of injection of iPSC-derived chondrocytes, chondrocytes were found on the top of the surface of the osteochondral defects with high expression of type I collagen.⁸⁷ Some somatic cells, like costal cartilages, have recently emerged as promising cells for TMJ tissue engineering due to their abundance of sources and superior ability in collagen synthesis compared with other cell types.⁸⁸ Vapniarsky et al utilized an aggregate redifferentiation method to produce costal chondrocytes to generate a cartilage sheet for TMJ disc repair in a minipig

model.²² The results showed that the tissue-engineered implants prevented the condylar head degeneration and successfully repaired the disc defects (Figure 3A).

5.2.2 Scaffolds in Tissue Engineering Strategies for TMJOA

Cell injection alone showed a low retention rate of transplanted cells in the joint due to severe anoikis and hypoxia.⁸⁹⁻⁹¹ Besides, the TMJ disc is under high mechanical stress and in motion. Therefore, scaffolds with adequate mechanical strength and long-term stability to ensure the high retention of transplanted cells are desired for cell-based tissue-engineered TMJ constructs.

Chitosan, alginate, collagen, and fibrin are the natural materials used in TMJ disc scaffolds. Chitosan shares structural similarities with numerous naturally occurring glycosaminoglycans found in cartilage.⁹² Alginate hydrogel characteristics can be designed to control stem cell fate and activity.⁹³ Bousnaki et al developed a kind of chitosan/alginate (Ch/Alg) scaffold embedded with DPSCs. They found the scaffold promoted DPSCs adhesion, viability, and proliferation, ultimately resulting in a storage modulus comparable to the human TMJ discs.⁹⁴ Collagen is a major component of native discs. In an investigation by Wang et al⁹⁵, the discs of rabbits regenerated and the joint's normal function was restored when the natural disc was removed by discectomy and subsequently replaced with reconstituted collagen templates embedded with ADMSCs. Moreover, Chen et al⁹⁶ used decellularized bovine distal femurs as scaffolds seeded with ADMSCs to regenerate the full-thickness cartilage in minipig joints after ramus-condyle unit ectomy (Figure 3B).

Polyglycolic acid (PGA)⁹⁷, polylactic acid (PLA)⁸², and polycaprolactone (PCL)⁹⁸ Poly-L-lactic-co-glycolic acid (PLGA)⁹⁹ are the common synthetic materials used in TMJ disc scaffolds due to their superior mechanical properties and controllability.¹⁰⁰ PGA is cytocompatible with TMJ disc-derived cells but it degrades fast.¹⁰¹ PLA has a longer degradation time. Aiming this, Ahtiainen et al implanted PLA-based scaffolds with ADMSCs to the rabbits' defective TMJ discs.⁸² They found PLA was sustained for 6 months without inflammatory symptoms and promoted cartilaginous tissue formation and bone regeneration in the TMJ. PCL possesses excellent mechanical strength but lacks bioactivity, which needs to be

modified to improve its biocompatibility¹⁰² for the reparation of cartilage defects.¹⁰³ PLGA has been licensed by the FDA for therapeutic use for cartilage regeneration. Uematsu et al. found that PLGA scaffolds can promote the colonization and proliferation of MSCs.⁹⁹ Tarafder et al designed a PCL scaffold embedded with PLGA microspheres and BMSCs, and noticed increased collagen synthesis and mechanical characteristics compared to the scaffold alone and enhanced healing of the perforated rabbits' TMJ discs after 4 weeks.⁹⁸

5.2.3 Growth Factors in Tissue Engineering Strategies for TMJOA

Soluble chemicals or growth factors are frequently elaborated to improve the biological characteristics of cells. For example, NEL-like molecule-1 (NELL-1) was employed to promote the chondrogenic differentiation of BMMSCs during pre-treatment.¹⁰⁴ Zhu et al observed that the NELL-1-modified BMMSCs showed more vigorous and well-organized fibrocartilage regeneration than those without NELL-1.¹⁰⁴ The transforming growth factors (TGF)-family^{82,98,105} and connective tissue growth factor (CTGF) can control the growth and function of the imported cells and newly formed tissues.¹⁰⁶ In a study by Tarafder et al⁹⁸, CTGF and TGF- β 3 were delivered with MSCs and PCL scaffolds. Compared to the control group, the experiment group significantly prevented arthritic changes of TMJ condyles, increased fibrocartilaginous recovery, and promoted multiphase fibrocartilaginous tissue formation in a disc perforation model. Moreover, platelet derivative growth factor (PDGF), basic fibroblast growth factor (bFGF), and insulin-like growth factor 1 (IGF-1)¹⁰⁷ were reported having the capacity to boost the proliferation rate of TMJ-disc derived cells, glycosaminoglycans formation, and collagen and hyaluronic acid synthesis¹⁰⁸ (**Table 3**) .

6. Limitations and Future Perspectives

TMJOA is difficult to be treated in the clinic due to the unique anatomy of TMJ and the complex etiology of this disease. Despite that different TMJOA models have been developed and specific mechanisms were revealed, each model has its advantages and shortcomings, and the majority of current models are single-factor models that do not adequately depict the etiology and progression of TMJOA. Therefore, it is feasible to create different multi-factor animal models for various situations, depending on the research purpose for investigating the neurobiological and molecular causes of TMJOA.⁵

Conventional detection methods, including imaging techniques, molecular techniques for detecting inflammatory cytokines, histochemical stainings and histomorphometry measures, are mostly used to diagnose TMJOA. These methods, however, are not specific and efficient in detecting TMJOA. For instance, inflammation also happens due to the infection in the temporal space, which is adjacent to the TMJ. Besides, the histological staining protocols vary in different labs, and the staining results are not objective, which may influence the diagnostic accuracy. Also, the conventional methods to diagnose TMJOA cannot be conducted in a high-throughput manner. In recent years, novel methods have been developed to detect diseases with high accuracy and specificity. For example, fluorine 18 fluorodeoxyglucose (FDG) is easily taken up by metabolically active inflammatory cells.¹¹⁶ Positron emission tomography (PET) identified increased uptake of FDG in periarticular lesions of patients with active symptoms, demonstrating it to be a more sensitive alternative to Micro-CT in detecting arthritic changes in the condyle. On the other hand, single-cell sequencing (SCS) can be used to obtain the information on genomes, transcriptomes, and epigenomes of individual cells, allowing researchers to identify new or rare cell types and compare healthy and disease-related tissues at single-cell resolution. Therefore, new diagnostic techniques such as SCS can be well-developed to diagnose TMJOA more specifically and efficiently.

According to this literature review, significant progress has been made in cell-based therapy of the TMJOA. However, several challenging issues have yet to be resolved. One of the most significant challenges in cell-based therapy is the insufficient understanding of therapeutic mechanisms and the difficulty in maintaining the precise and prolonged persistence of implanted cells in the TMJ. More recent studies have noted that secretomes by implanted cells⁸⁹, particularly exosomes, play a significant role in cell-based regenerative constructs. Several reports suggest that exosomes produced from stem cells trigger the migration of progenitor cells, promote cartilage and bone regeneration, and pain reduction in TMJOA animal models.⁵⁵ Therefore, exosomes with advanced delivery systems can be a more potential and novel strategy for treating TMJOA.

The other concern of tissue-engineering-based therapy is the low efficiency of fibrocartilage restoration on the articulating surfaces and implant displacement due to the

complex structure of TMJ. Aiming this, three-dimensional (3D) bioprinting is a potential and unique fabrication approach for accurately positioning biologics, such as living cells and extracellular matrix ECM components, in a specified 3D hierarchical order to build artificial multicellular TMJs with high efficiency. 3D bioprinting used in TMJOA treatment is still rarely reported, which is worth to be attempted in the future.

7. Conclusion

Appropriate animal models can help researchers better understand the pathophysiology of TMJOA and find therapeutic options. The currently developed models and diagnostic methods have their benefits and limitations. In the future, it is necessary to induce different multi-factor animal models and clinical symptoms relevant models with advanced diagnostic techniques of higher accuracy and specificity to depict the neurobiological and molecular causes of TMJOA. The limitations of current TMJOA treatments inspire us to develop new drug delivery systems and tissue engineering strategies to treat TMJOA. However, several challenging issues have yet to be resolved. Increasing the effectiveness of cell-based therapy and tissue engineering products and understanding more about the therapeutic mechanisms will be the focus of future investigations on the treatment of TMJOA.

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Disclosure Statement

The authors declare that no conflict of interests exist.

Authors' Contributions

Dongyun Wang: conceptualization, article writing, and funding support. Yajie Qi: literature collection, image processing, and article writing. Zhubing Wang and Anyun Guo: literature collection and image processing. Yingxin Xu and Yang Zhang: conceptualization, literature collection, article writing, and funding support.

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Table 1. Reported animal models of TMJOA

Types	Methods	Species	Pathological manifestations	Reference
Naturally Occurring Models	Using aging mice	C57BL/6N Mice	Cartilage destruction, proliferative layer exposed	Zheng et al. ^[10] & Ukita et al. ^[11]
Chemical Induction	Injection of CFA	Sprague–Dawley Rats	Reduced thickness of cartilage, induced chondrocytes death, and decreased proteoglycan in the articular cartilage	Cui et al. ^[12]
	Injection of MIA	Sprague–Dawley Rats	Decreased the number of chondrocytes and percentage areas of proteoglycan, subchondral bone remodeling, enhanced osteophyte, increased number of gliosis	Jiang et al. ^[13] &Wang et al. ^[14]
	Injection of VEGF	Sprague-Dawley Rats	Cartilage and subchondral bone destruction, increased the number of osteoclasts in the subchondral bone	Shen et al. ^[15]
	Injection of collagenase	C57BL/6J Mice	Induced irregular condylar surface and increased expression of COX-2, MMP1, and MMP13	Li et al. ^[16]
	Injection of formalin	Wistar Rats	Induced synovial membrane hyperplasia, inflammatory reaction in the condylar neck, and atrophy of the fibrocartilage	Goulart et al. ^[17]
	Disc perforation	New Zealand Rabbits	Cartilage degeneration and heterotopic ossification within TMJ discs	Embree et al. ^[18]
Surgical Operation	Joint disc deformation	New Zealand	Synovial hyperplasia, joint disc deformation	Liu et al. ^[19]

		Rabbits		
	Anterior disc displacement	Japanese Rabbits	Enhanced chondrocyte apoptosis, decreased cell density in cartilage	Xu et al. ^[20]
	Partial discectomy	C57BL Mice	Reduced proteoglycan in articular cartilage and bone loss	Cohen et al. ^[21]
	Condylectomy	Minipig	Induced subchondral cysts and large osteophytes formation with the extensive irregular cartilage surface	Vapniarsky et al. ^[22]
	Condylectomy	C57BL/6 Mice	Discontinued cartilage and bone loss, without cartilage regeneration and extracellular matrix formation	Gomez et al. ^[23]
	Unilateral anterior cross	Sprague–Dawley Rats	Induced chondrocyte death, aberrant mineral deposition in cartilage	Zhang et al. ^[24]
	Unilateral anterior cross	C57BL/6 Mice	Decreased the density of chondrocyte, thickness of cartilage, and percentage of the proteoglycan areas	Sun et al. ^[25]
Mechanical Stimulation	Abnormal occlusion	Sprague–Dawley Rats	Condylar cartilage degradation, increased chondrocyte death in mandibular condyle cartilage	Jiao et al. ^[26]
	Bonding a stainless wire on the molars	C57BL/6 Mice	Condylar cartilage surface resorption, decreased proteoglycan levels, increased expression of MMP13	Matias et al. ^[27]
	Build-up of resin on the molars	C57BL/6 Mice	Decreased thickness of condylar cartilage, increased osteoclastic activity, and inflammatory responses	Jung et al. ^[28]
Gene	Shox2-depleted	C57BL/6J	TMJ defects, premature articular disc erosion	Liang et al. ^[29]

Modification	Mice		
<i>Prg4-null</i>	C57BL/6J <i>Mice</i>	Abnormal cartilage, osteophytes formation in glenoid fossa	Bechtold et al. ^[30]
DDR-1 knockout	C57BL/6 <i>Mice</i>	Cartilage surface fissures, proteoglycan degradation, chondrocyte cluster development	Schminke et al. ^[31]

Abbreviations: CFA, complete Freund's adjuvant; MIA, monosodium iodine acetate; BSA, bovine serum albumin; VEGF, vascular endothelial growth factor; IL-1, interleukin-1; IL-6, interleukin-6; TNF, tumor necrosis factor; COX, cyclooxygenase; MMP, matrix metalloproteinase; OPN, osteopontin; RANKL, receptor activator of NF-κB ligand; Shox2, short stature homeobox 2; Prg4, Proteoglycan4; DDR, discoidin domain receptor.

Table 2. Comparison of established TMJOA animal models by different methods

Methods	Pathogenesis of the TMJOA	Advantages	Disadvantages
Naturally Occurring	Aging	Simple, highly reproducible	Time-consuming
Chemical Induction	Inflammatory stimulation	Simple operation, controllable damage, reproducible symptoms	Short maintaining time, cannot closely resemble the degeneration process
Surgical Operation	Trauma	Allowing for a comprehensive evaluation of the joint	Restricted in large animal models
Mechanical Stimulation	Excessive mechanical stress on TMJ	Efficient, cost-efficient, highly reproducible, minor damage to animals	Occurrence of adverse trauma or joint dislocation
Gene Modification	Genetic factors	Reflecting the course of the disease, investigating specific factors	High cost, limited to small animals

Table 3. Tissue Engineering Therapy for the treatment of TMJOA

Delivery system	Cells	Drugs	Animals	Results	Reference
Chitosan hydrogels, ADH, HA-ALD	-	HA	Rat	Protected cartilage degeneration, alleviated pain, inhibited the expression of inflammatory cytokines after 12 weeks injection	Mou et al. ^[109]
PLGA	-	HA, parecoxib	Rat	Declined the TNF- α , IL-1 β , and COX-2 expression in articular synovial cells, relieved of inflammation and nociceptive reactions	Zhu et al. ^[110]
PLGA	-	PGJ2	Rat	Decreased the release of IL-1 β and hyperalgesic response in TMJOA models	Clemente-Napimoga et al. ^[111]
CP, CCT, Pluronic F68	-	Naproxen	Rat	Reduced the migration of leukocytes and levels of IL-1 β and TNF- α	Guilherme et al. ^[72]
MSN-CC-PEI	-	HAS2	Rat	Promoted endogenous HA production, reduced synovial inflammation for 3 weeks, and enhanced bone repair in a rat OA bone defect model	Li et al. ^[73]
PL407	-	DK, HA	Rat	Regenerated cartilage, controlled D.K. release for 4 days	Hanafy and El-Ganainy Abdalla et al. ^[74]

PL407	-	15d-PGJ ₂	Rat	Relieved pain at lower concentrations in a sustained manner, reduced levels of pro-inflammatory cytokines	Abdalla et al. ^[112]	40
Pluronic F-127	Chondrocytes	-	Goat	Neocondylar surface covered with fibrocartilaginous tissue, the condylar bone and cartilage rehabilitation and regeneration in the TMJOA induced by condylar defection	Yu et al. ^[113]	
-	Chondrocytes, BMSCs	-	Goat	Formed the cartilage-like tissues, increased the type II collagen in the condylar cartilage defects	Sun et al. ^[114]	
-	DPSCs	-	Rat	Relieved inflammatory pain, decreased cartilage degradation, induced bone regeneration, and downregulated the expression levels of MMP3 and TNF- α	Cui et al. ^[12]	
SHED-CM	-	-	Mice	Inhibited temporal muscle inflammation and cartilage devastation, induced condylar cartilage, and subchondral bone regeneration	Ogasawara et al. ^[55]	
PRP	BMMSCs	-	Mice	Newly cartilage and bone covered at the top of the condyle defect with extracellular matrix deposition	Gomez et al. ^[23]	
PLGA	BMMSCs	NELL-1	Goat	Promoted the chondrogenic differentiation of BMMSCs, better-organized repair of fibrocartilage regeneration	Zhu et al. ^[104]	

PLGA, PCL	BMMSCs	CTGF, TGF- β 3	Rabbit	Prevented arthritic symptoms on TMJ condyles, promoted fibrocartilaginous recovery and multiphase fibrocartilaginous tissues formation	Tarafder et al. ^[98]	41
Decellularized bovine distal femurs	ADMSCs	TGF- β	Minipig	The tissue-engineered ramus-condyle unit kept its intended anatomical structure, regenerated the full-thickness cartilage over the underlying bone	Chen et al. ^[96]	
-	Costal chondrocytes	TGF- β 1	Minipig	Repaired the disc defects and restored the disc integrity	Vapniarsky et al. ^[22]	
PCL, gelatin	IPSCs	-	Rabbit	Increased cartilage-specific gene expression and protein levels, subchondral bone regeneration	Liu et al. ^[103]	
-	-	Exosomes	Rat	Triggered migration of progenitor cells, significant cartilage and bone regeneration, and pain reduction	Zhang et al. ^[115]	

Abbreviations: ADH, adipic acid dihydrazide; HA-ALD, hyaluronic acid–aldehyde; PLGA, poly (lactic-co-glycolic acid); PEI, poly (ethylenimine); I.L., interleukin; 15d-PGJ2, 15-deoxy- Δ 12,14-prostaglandin J2; PCL, polycaprolactone; PLA, polylactic acid; CP, cetyl palmitate; CCT, capric/caprylic triglycerides; MSN-CC, mesoporous silica nanoparticles with a core-cone structure; HAS2, hyaluronan synthase 2; PL, poloxamer; DK, diclofenac potassium; 15d-PGJ₂, 15-deoxy- Δ 12,14-prostaglandin J₂; 3D, three dimensional 3D; PCL, polycaprolactone; BMMSCs, bone marrow-derived stem cells; DPSCs, dental pulp stem cells; SHED-CM, conditioned media from human exfoliated deciduous teeth stem cells; PRP, platelet-rich plasma; NELL-1, NEL-like molecule-1; TGF, transforming growth factors; CTGF, connective tissue growth factor; ADMSCs, Adipose-derived stem cells ; BMP-2, bone morphogenetic protein 2; ECM, decellularized extracellular matrix.

Figure legend:

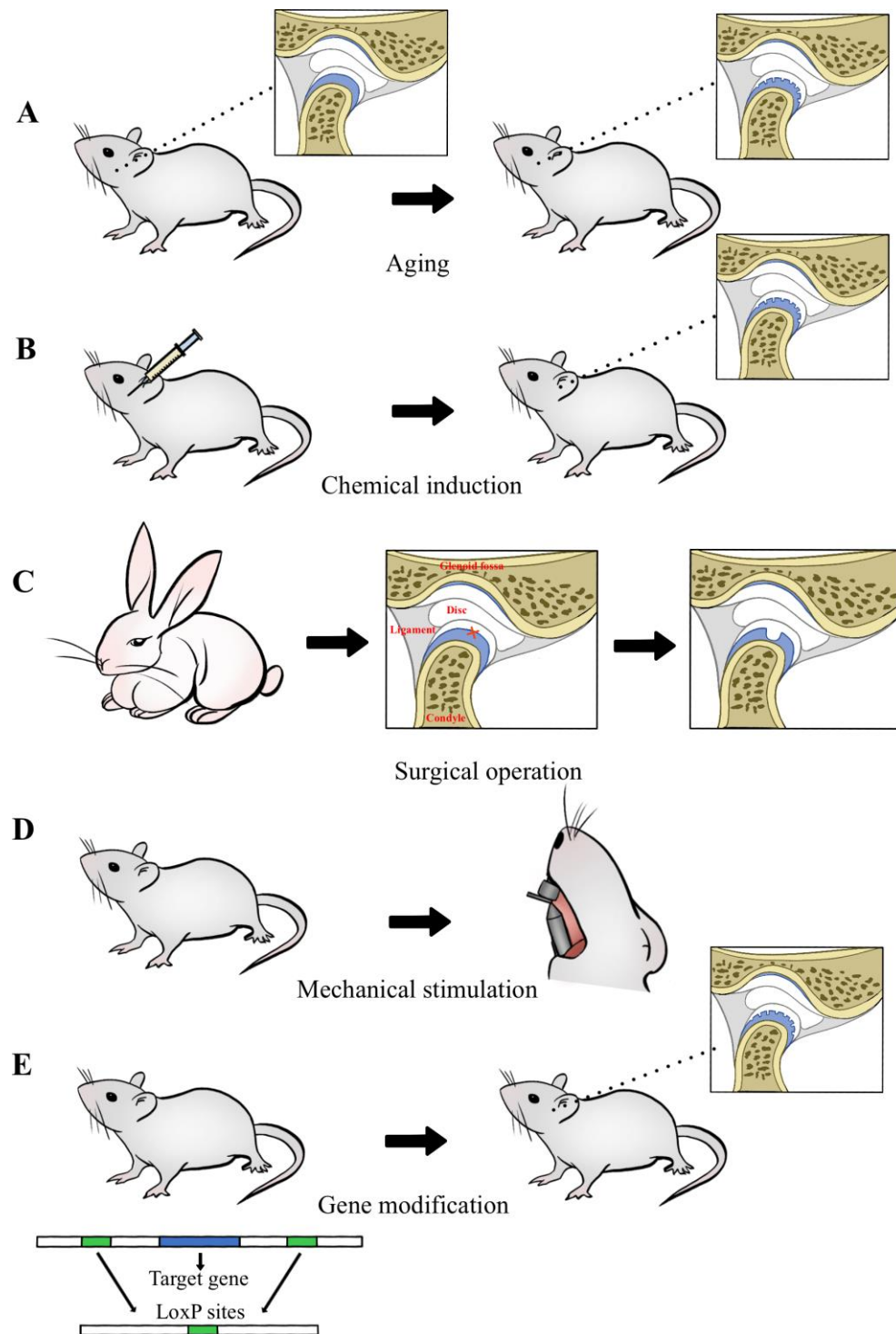


Figure 1: Different methods to establish TMJOA animal models.

(A) Naturally occurring models of TMJOA. (B) Chemical induction of TMJOA. (C) Surgical operation of TMJOA. (D) Mechanical stimulation of TMJOA. (E) Gene modification of TMJOA.

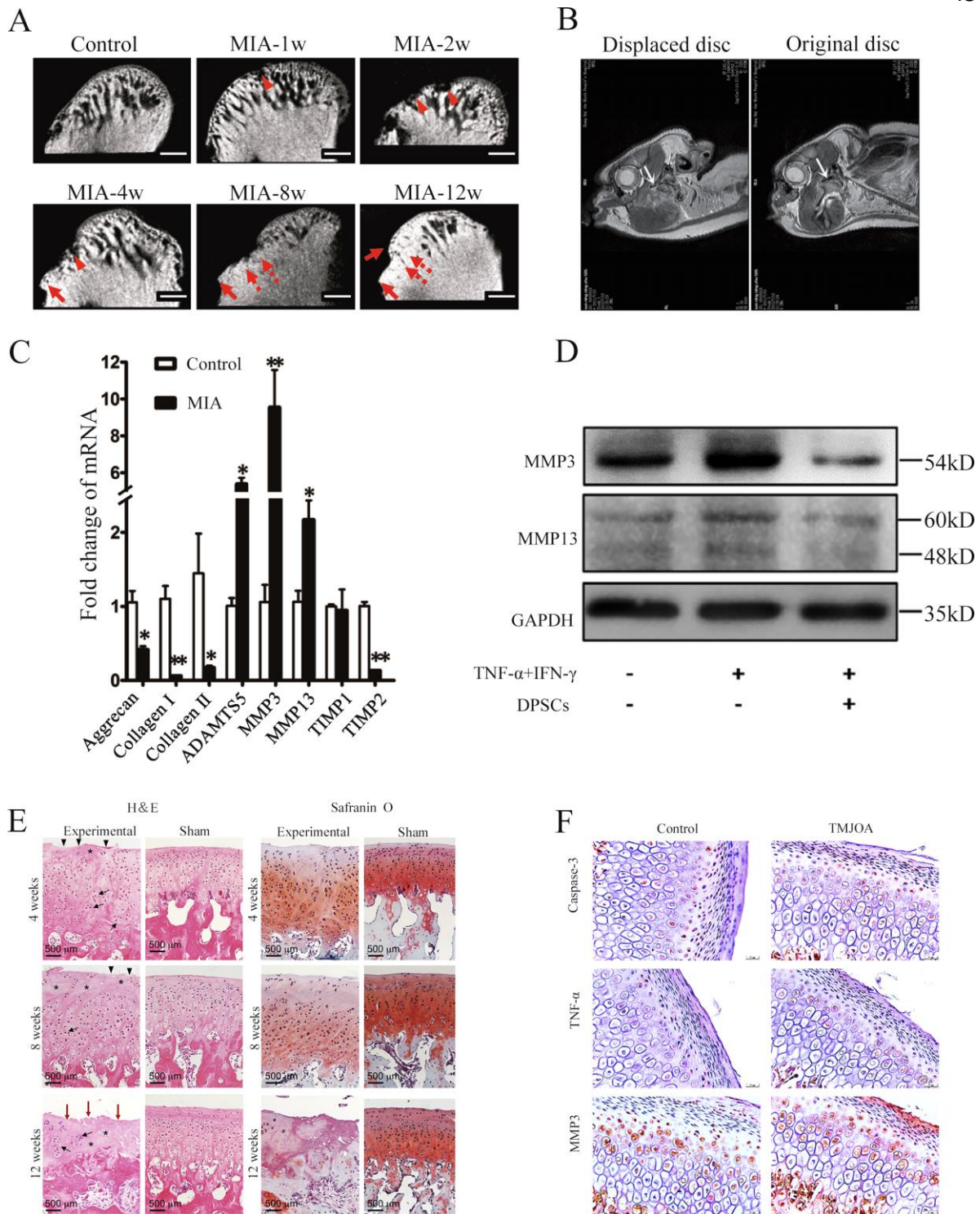


Figure 2: Different approaches to diagnose TMJOA in animal studies. (A) Representative images of condyle erosions of TMJOA rat by Micro-CT scanning^[14]. (B) Representative images of the displaced disc in condylar fracture induction TMJOA minipig by MRI^[38]. (C) Gene expression and (D) protein content of matrix degrading proteases were significantly upregulated in TMJOA rat evaluated by RT-PCR^[14] and western Blot^[12], respectively. (E)

Histological sections of condyles in disc perforation of TMJOA rabbit by H&E and safranin O staining^[18]. (F) Apoptosis-related cytokines **of condyles** was detected **in** TMJOA rat by histochemical stainings^[61]. Adapted with permission.

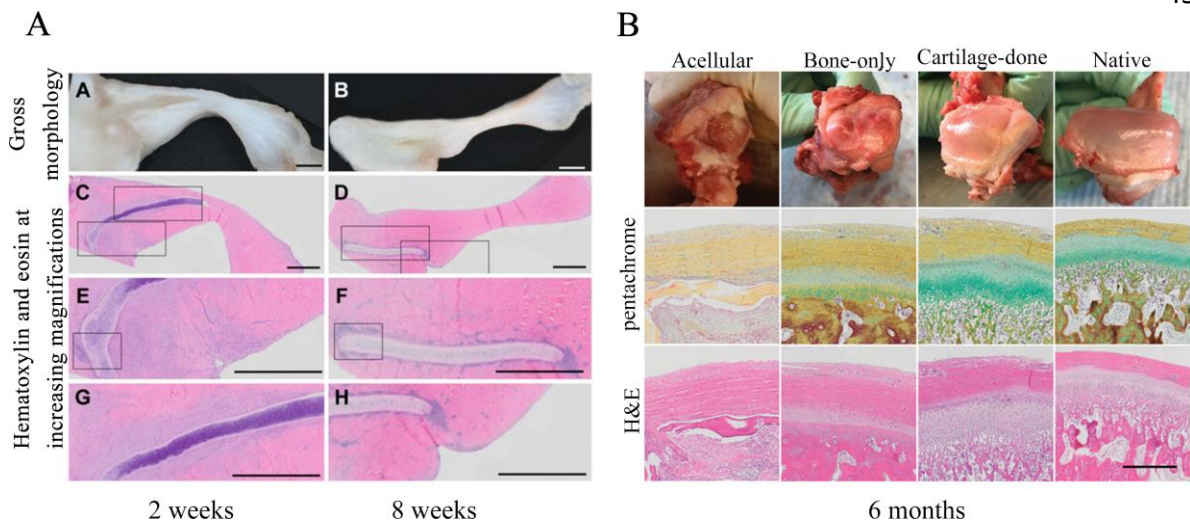


Figure 3: Tissue Engineering Strategies for TMJOA. (A) Tissue-engineered TMJ discs were engineered with allogeneic costal chondrocytes and applied in the orthotopic TMJ of minipigs. These implants restored disc integrity by inducing complete defect closure, formed repair tissue, and promoted intralaminar fusion compared to empty controls^[22]. (B) Decellularized bovine distal femurs and ADMSCs were utilized to construct tissue-engineered TMJ discs to replace the original mandibular ramus-condyle unit of minipigs. **These** implants restored cartilage integrity by regenerating the full-thickness, stratified and robust cartilage over the underlying bone compared to empty controls^[96]. Adapted with permission.